Electric Vehicle Charging Pattern Analysis Report

❖ Objective:

- The objective of this project is to design and implement an end-to-end data pipeline and interactive dashboard to analyze electric vehicle charging patterns. The project uses MySQL for data processing and Power BI for visualization to identify trends in charging behaviour, efficiency and station utilization. The insights support infrastructure planning, resource optimization and the wider adoption of electric vehicles.

❖ Dataset Overview:

- Source: Kaggle – Electric Vehicle Charging Dataset (CSV format)
- Time Period: January 2024 to February 2024
- Total Records: 1320 rows
- Key Columns:

- User ID
- Vehicle ID
- Vehicle Model
- User Type
- Charging Station ID
- Charging Start Time
- Charging End Time
- Charging Duration (mins)
- Charging Duration (hours)
- Energy Consumed (kWh)
- Charging Rate (kW)
- Distance Driven (since last charge) (km)

❖ Tools & Technologies:

- Dataset: CSV file containing EV charging session data
- Database: MySQL Workbench for importing data, creating schema, cleaning and validation using SQL
- Data Visualization: Microsoft Power BI for data modelling, DAX calculations and interactive dashboard creation.
- Languages: SQL for queries and transformations, DAX for measures and calculated columns.

❖ Database Schema:

- Database Creation:

```sql
CREATE DATABASE EV_ChargingDB;
USE EV_ChargingDB;

```

- Table Name: EV_Charging
- Columns:

- User ID (VARCHAR)
- Vehicle Model (VARCHAR)
- Battery Capacity (kWh) (DECIMAL)
- Charging Station ID (VARCHAR)
- Charging Station Location (VARCHAR)
- Charging Start Time (DATETIME)
- Charging End Time (DATETIME)
- Energy Consumed (kWh) (DECIMAL)
- Charging Duration (hours) (DECIMAL)
- Charging Rate (kW) (DECIMAL)
- Charging Cost (USD) (DECIMAL)
- Time of Day (VARCHAR)
- Day of Week (VARCHAR)
- State of Charge (Start %) (DECIMAL)
- State of Charge (End %) (DECIMAL)
- Distance Driven (since last charge) (km) (DECIMAL)
- Temperature (°C) (DECIMAL)
- Vehicle Age (years) (DECIMAL)
- Charger Type (VARCHAR)
- User Type (VARCHAR)

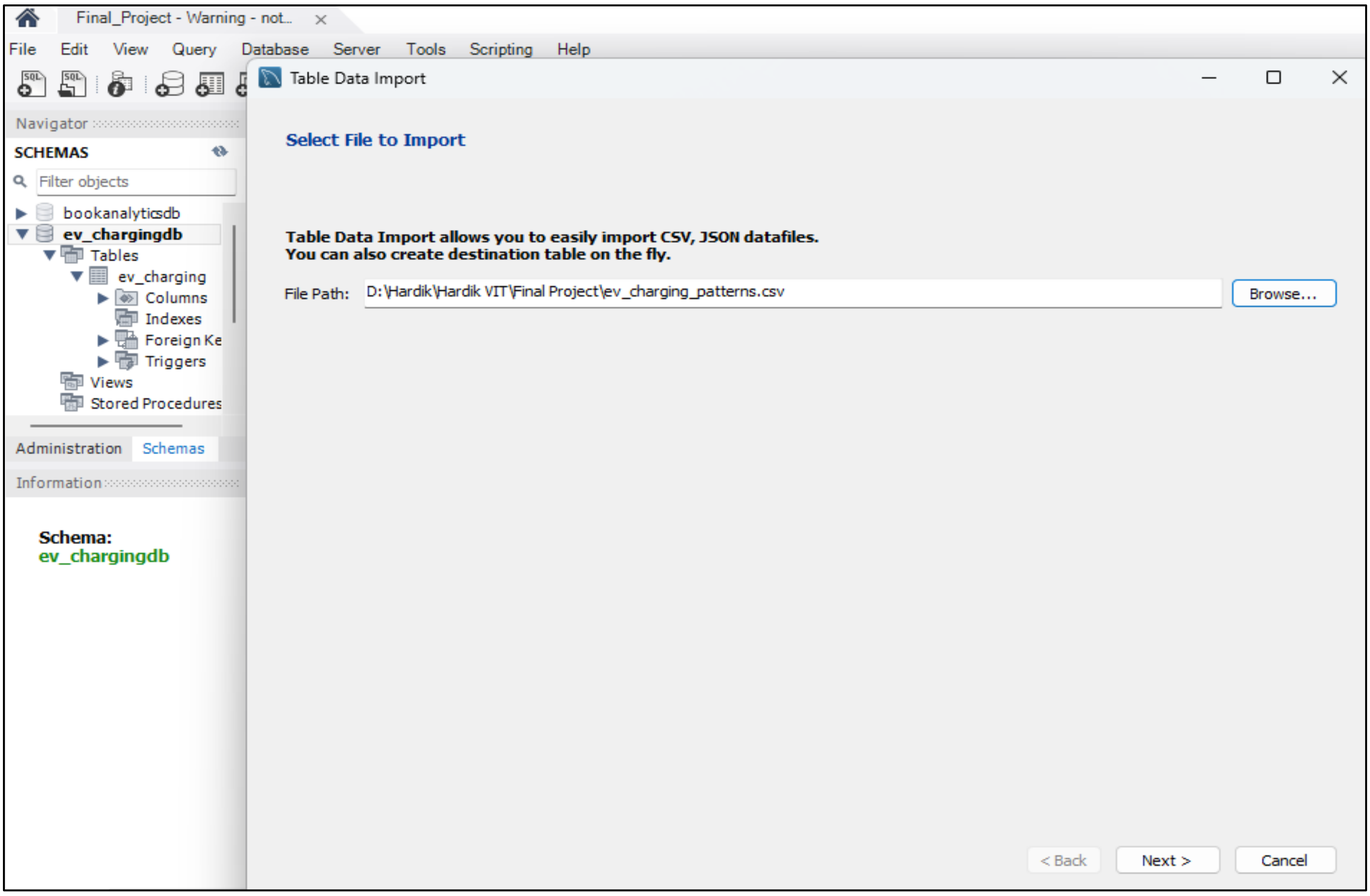
❖ **<u>SQL Implementation:</u>**

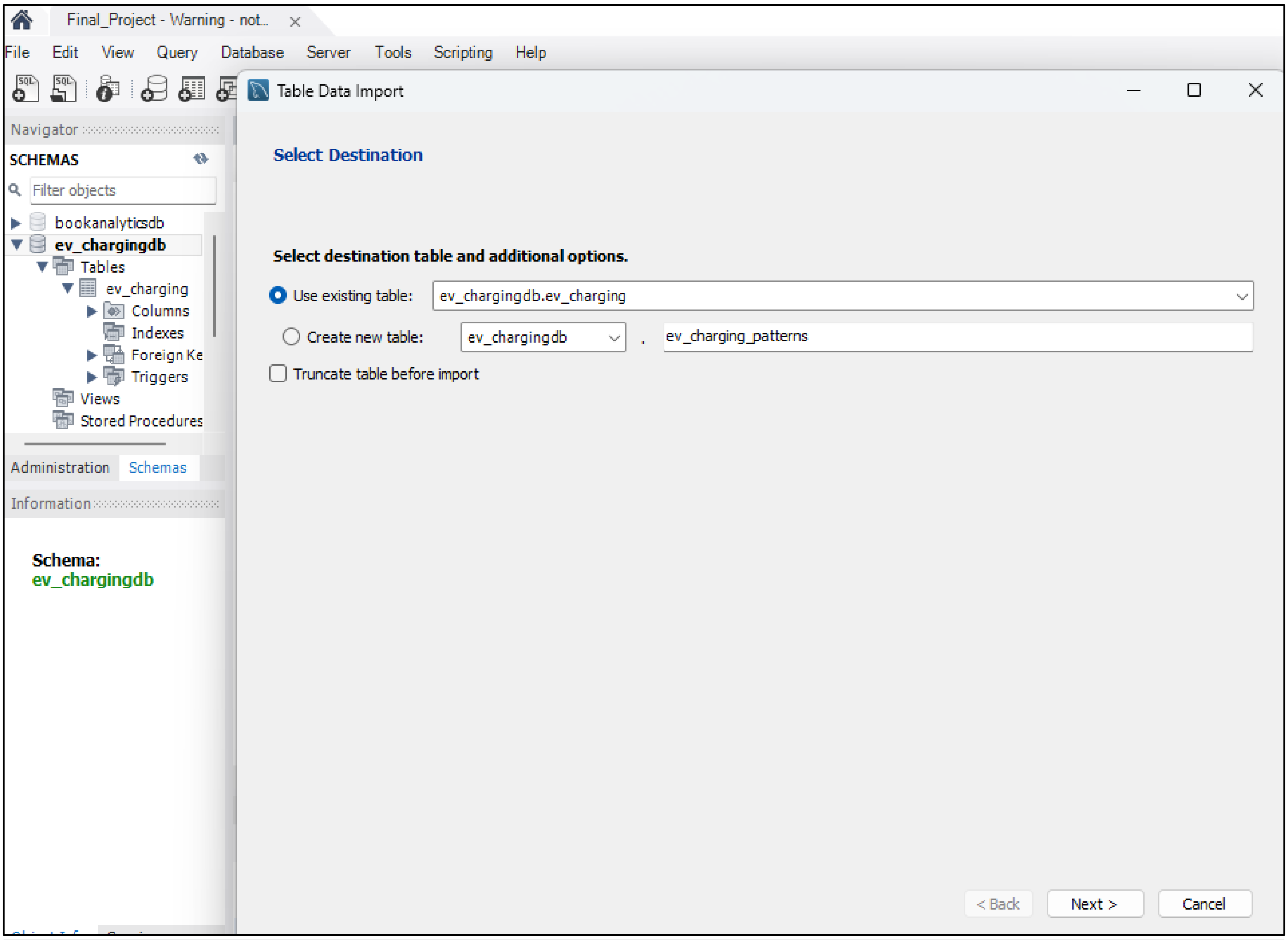
➢ The dataset was imported into MySQL Workbench for cleaning, transformation and validation. The following steps were applied:

- Table Creation: Created the table EV_Charging with all relevant columns including numeric, text and datetime fields.

```sql
CREATE TABLE EV_Charging(
 `User ID` VARCHAR(100),
  `Vehicle Model` VARCHAR(150),
  `Battery Capacity (kWh)` DECIMAL(6,2),
  `Charging Station ID` VARCHAR(100),
  `Charging Station Location` VARCHAR(255),
  `Charging Start Time` VARCHAR(40),    -- will later convert to DATETIME
  `Charging End Time` VARCHAR(40),      -- will later convert to DATETIME
  `Energy Consumed (kWh)` DECIMAL(8,3),
  `Charging Duration (hours)` DECIMAL(6,3),
  `Charging Rate (kW)` DECIMAL(6,3),
  `Charging Cost (USD)` DECIMAL(10,2),
  `Time of Day` VARCHAR(50),
  `Day of Week` VARCHAR(30),
  `State of Charge (Start %)` DECIMAL(5,2), -- percentage
  `State of Charge (End %)` DECIMAL(5,2),   -- percentage
  `Distance Driven (since last charge) (km)` DECIMAL(9,2),
  `Temperature (°C)` DECIMAL(5,2),
  `Vehicle Age (years)` DECIMAL(5,2),
  `Charger Type` VARCHAR(50),
  `User Type` VARCHAR(50));
```

- Data Import: Imported the CSV file using the Table Data Import Wizard in MySQL Workbench.

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Navigator
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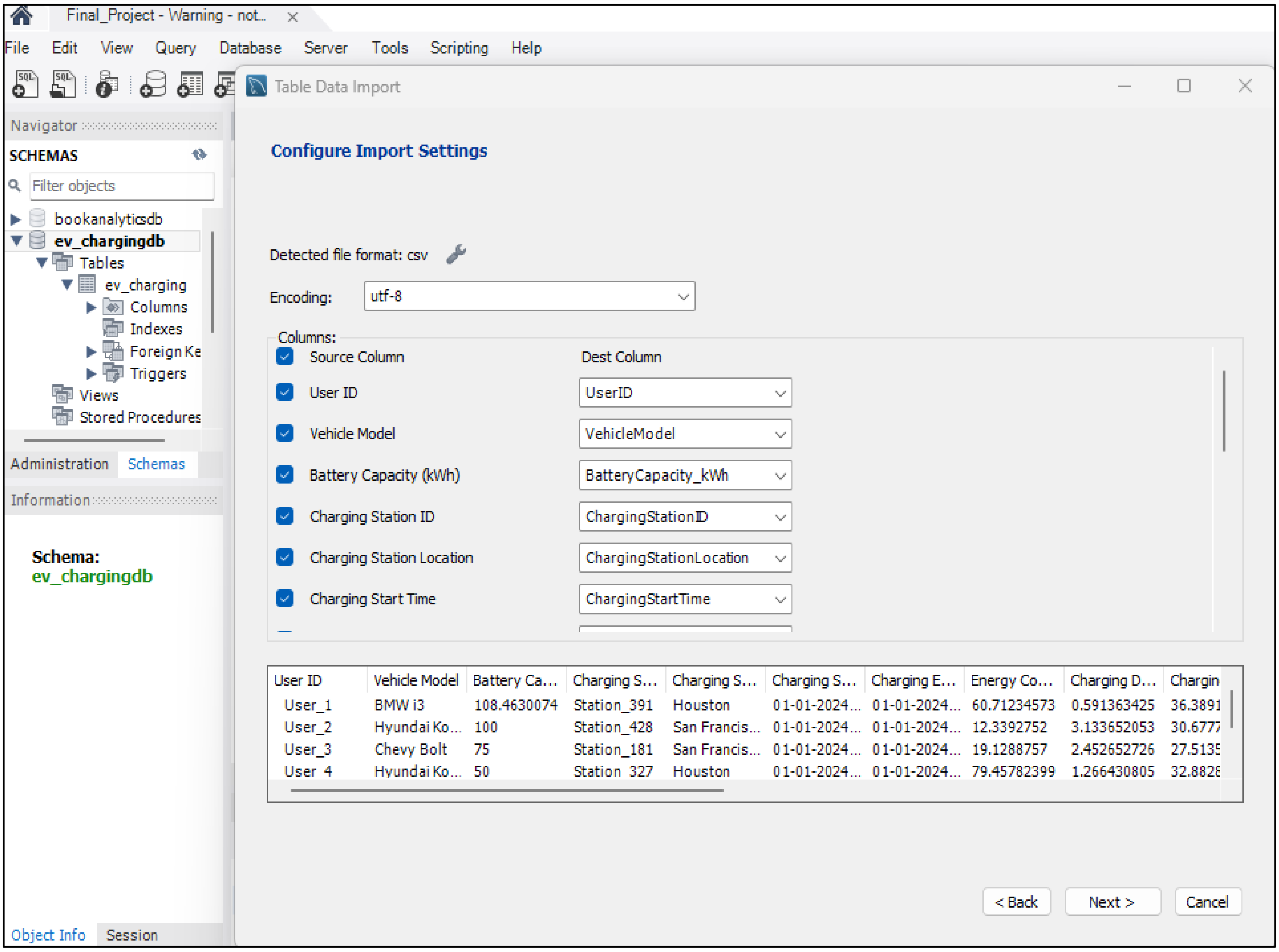
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Administration Schemas
Information
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ev_chargingdb
Configure Import Settings
Detected file format: csv
Encoding:
utf-8
Columns:
Source Column
Dest Column
User ID
UserID
Vehicle Model
VehicleModel
Battery Capacity (kWh)
BatteryCapacity_kWh
Charging Station ID
ChargingStationID
Charging Station Location
ChargingStationLocation
Charging Start Time
ChargingStartTime
User ID Vehicle Model Battery Ca... Charging S... Charging S... Charging S... Charging E... Energy Co... Charging D... Chargin
User_1 BMW i3 108.4630074 Station_391 Houston 01-01-2024... 01-01-2024... 60.71234573 0.591363425
User_2 Hyundai Ko... 100 Station_428 San Francis... 01-01-2024... 01-01-2024... 12.3392752 3.133652053
User_3 Chevy Bolt 75 Station_181 San Francis... 01-01-2024... 01-01-2024... 19.1288757 2.452652726
User_4 Hyundai Ko... 50 Station_327 Houston 01-01-2024... 01-01-2024... 79.45782399 1.266430805
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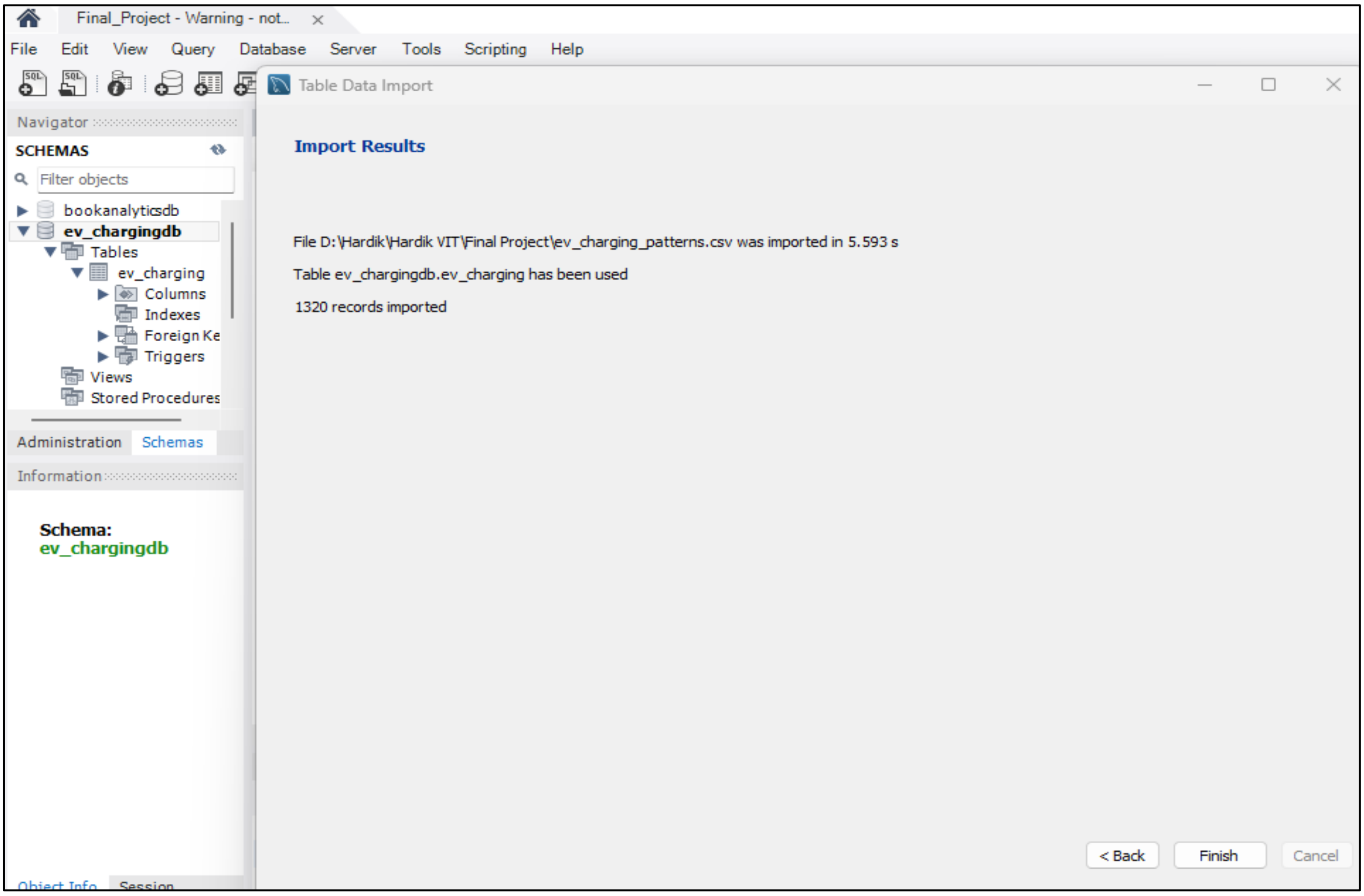


- Verifying Import: Testing sample queries in MySQL Workbench.

```
27 • SELECT * FROM EV_Charging LIMIT 10;
28
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content: | Fetch

id	UserID	VehicleModel	BatteryCapacity_kWh	ChargingStationID	ChargingStationLocation	ChargingStartTime
1	BMW i3	108.4630074	0.00	Houston	01-01-2024 00:00	01-01-2024 00:39
2	Hyundai Kona	100	0.00	San Francisco	01-01-2024 01:00	01-01-2024 03:01
3	Chevy Bolt	75	0.00	San Francisco	01-01-2024 02:00	01-01-2024 04:48
4	Hyundai Kona	50	0.00	Houston	01-01-2024 03:00	01-01-2024 06:42
5	Hyundai Kona	50	0.00	Los Angeles	01-01-2024 04:00	01-01-2024 05:46
6	Nissan Leaf	50	0.00	San Francisco	01-01-2024 05:00	01-01-2024 07:10
7	Chevy Bolt	85	0.00	Houston	01-01-2024 06:00	01-01-2024 07:53
8	Chevy Bolt	75	0.00	Los Angeles	01-01-2024 07:00	01-01-2024 10:42
9	Chevy Bolt	62	0.00	Los Angeles	01-01-2024 08:00	01-01-2024 09:21
10	Hyundai Kona	50	0.00	Chicago	01-01-2024 09:00	01-01-2024 12:44
NULL	NULL	NULL	NULL	NULL	NULL	NULL

```
26 • SELECT COUNT(*) FROM EV_Charging;
27 • SELECT * FROM EV_Charging LIMIT 10;
```

Result Grid | Filter Rows: | Expo

COUNT(*)
1320

❖ Data Cleaning and Transformation:

➢ The following steps were executed in MySQL Workbench to clean the dataset and prepare it for analysis.

1) Checking sum of null or 0.0 values in columns: To identify rows with invalid or placeholder values that require cleaning.

```
SELECT          -- Check for NULLs in each column
    SUM(`Energy Consumed (kWh)` IS NULL OR `Energy Consumed (kWh)` = 0.0) AS Issue_Energy,
    SUM(`Charging Rate (kW)` IS NULL OR `Charging Rate (kW)` = 0.0) AS Issue_Rate,
    SUM(`Distance Driven (since last charge) (km)` IS NULL OR `Distance Driven (since last char
FROM EV_Charging;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Issue_Energy	Issue_Rate	Issue_Distance
66	66	66

2) Finding averages: To compute realistic replacement values for invalid 0.0 entries in numeric columns.

```
SELECT          -- Find Averages
    AVG(`Energy Consumed (kWh)`) AS Avg_Energy,
    AVG(`Charging Rate (kW)`) AS Avg_Rate
FROM EV_Charging
WHERE `Energy Consumed (kWh)` <> 0.0
  AND `Charging Rate (kW)` <> 0.0;
```

Result Grid | Filter Rows: | Export: | Wrap Cell

Avg_Energy	Avg_Rate
42.8882477	25.8679160

3) Replacing zeros with average values: To impute reasonable values for Energy Consumed (kWh) and Charging Rate (kW) while preserving all rows. Approach was update rows where the column equals 0.0 with the column average. Because Workbench runs in safe update mode, safe_updates was temporarily disabled for the bulk update (eg: SET SQL_SAFE_UPDATES = 0/1;).

```
UPDATE EV_Charging    -- Replace 0.0 with average in Energy Consumed (kWh)
SET `Energy Consumed (kWh)` = 42.8882477
WHERE `Energy Consumed (kWh)` = 0.0;

UPDATE EV_Charging    -- Replace 0.0 with average in Charging Rate (kW)
SET `Charging Rate (kW)` = 25.8679160
WHERE `Charging Rate (kW)` = 0.0;
```

4) Validation after update: To confirm that no zero entries remain for the cleaned columns.

```sql
SELECT                          -- Validation after update
    SUM(`Energy Consumed (kWh)` = 0.0) AS Zero_Energy,
    SUM(`Charging Rate (kW)` = 0.0) AS Zero_Rate
FROM EV_Charging;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Zero_Energy	Zero_Rate
0	0

5) Adding new columns in table: To create proper typed columns to hold converted timestamps and derived values.

```sql
ALTER TABLE EV_Charging  -- new columns in table
ADD COLUMN ChargingStart DATETIME,
ADD COLUMN ChargingEnd DATETIME;
```

Output

Action Output

#	Time	Action	Message
9	15:38:03	SELECT SUM(`Energy Consumed (kWh)` = 0.0) ...	1 row(s) returned
10	15:51:08	ALTER TABLE EV_Charging -- new columns in ta...	0 row(s) affected Records: 0 Duplicates:

6) Converted existing text into datetime format: To convert original text timestamp columns into proper DATETIME fields for accurate time calculations.

```sql
UPDATE EV_Charging          -- Converted existing text into datetime format
SET ChargingStart = STR_TO_DATE(`Charging Start Time`, '%d-%m-%Y %H:%i:%s'),
    ChargingEnd   = STR_TO_DATE(`Charging End Time`, '%d-%m-%Y %H:%i:%s');
```

Output

Action Output

#	Time	Action	Message
13	15:55:19	UPDATE EV_Charging -- Converted existing te...	1320 row(s) affected Rows matched: 1320 Chang...

7) Verify conversion: To inspect sample rows to ensure datetime conversion worked correctly.

```sql
SELECT `Charging Start Time`, ChargingStart,    -- Verify conversion
       `Charging End Time`, ChargingEnd
FROM EV_Charging
LIMIT 10;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

Charging Start Time	ChargingStart	Charging End Time	ChargingEnd
01-01-2024 00:00	2024-01-01 00:00:00	01-01-2024 00:39	2024-01-01 00:39:00
01-01-2024 01:00	2024-01-01 01:00:00	01-01-2024 03:01	2024-01-01 03:01:00
01-01-2024 02:00	2024-01-01 02:00:00	01-01-2024 04:48	2024-01-01 04:48:00
01-01-2024 03:00	2024-01-01 03:00:00	01-01-2024 06:42	2024-01-01 06:42:00
01-01-2024 04:00	2024-01-01 04:00:00	01-01-2024 05:46	2024-01-01 05:46:00
01-01-2024 05:00	2024-01-01 05:00:00	01-01-2024 07:10	2024-01-01 07:10:00
01-01-2024 06:00	2024-01-01 06:00:00	01-01-2024 07:53	2024-01-01 07:53:00
01-01-2024 07:00	2024-01-01 07:00:00	01-01-2024 10:42	2024-01-01 10:42:00
01-01-2024 08:00	2024-01-01 08:00:00	01-01-2024 09:21	2024-01-01 09:21:00
01-01-2024 09:00	2024-01-01 09:00:00	01-01-2024 12:44	2024-01-01 12:44:00

8) Dropping old columns: To remove the original text timestamp columns after successful conversion to reduce confusion and avoid duplicate fields.

```sql
ALTER TABLE EV_Charging    -- Dropping old columns
DROP COLUMN `Charging Start Time`,
DROP COLUMN `Charging End Time`;
```

Output

Action Output

	#	Time	Action	Message
✔	15	15:56:39	SELECT `Charging Start Time`, ChargingStart, ...	10 row(s) r
✔	16	15:58:35	ALTER TABLE EV_Charging -- Dropping old colu...	0 row(s) af

9) New calculated column in SQL: To add a numeric column storing session duration in minutes for easier analysis and for use in Power BI.

```sql
ALTER TABLE EV_Charging    -- new calculated column
ADD COLUMN ChargingDuration_mins INT;
```

```sql
UPDATE EV_Charging        -- calculates the difference in minutes
SET ChargingDuration_mins = TIMESTAMPDIFF(MINUTE, ChargingStart, ChargingEnd);
```

10) Verifying calculated column: To confirm the derived duration matches the difference between start and end datetimes.

```sql
SELECT ChargingStart, ChargingEnd, ChargingDuration_mins -- Verifying
FROM EV_Charging
LIMIT 10;
```

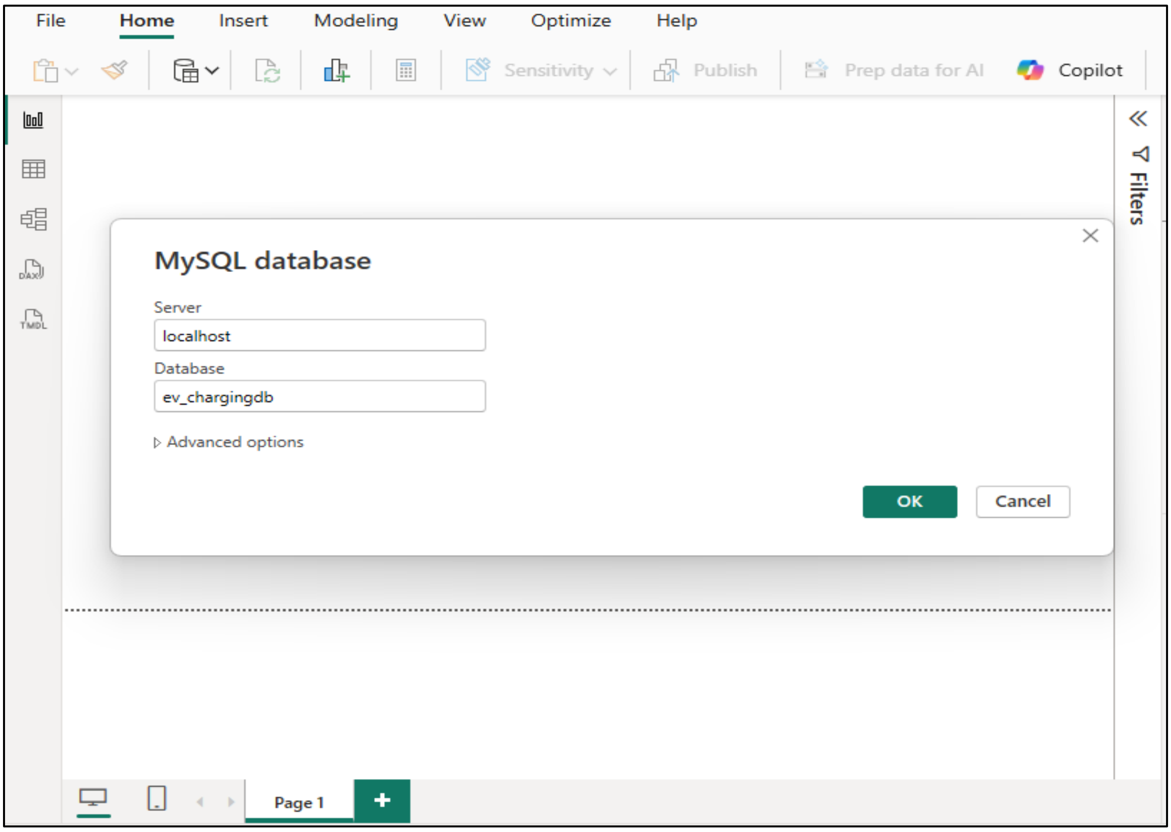
Result Grid | Filter Rows: | Export: | Wrap Cell Content: | Fetch rows:

ChargingStart	ChargingEnd	ChargingDuration_mins
2024-01-01 00:00:00	2024-01-01 00:39:00	39
2024-01-01 01:00:00	2024-01-01 03:01:00	121
2024-01-01 02:00:00	2024-01-01 04:48:00	168
2024-01-01 03:00:00	2024-01-01 06:42:00	222
2024-01-01 04:00:00	2024-01-01 05:46:00	106
2024-01-01 05:00:00	2024-01-01 07:10:00	130
2024-01-01 06:00:00	2024-01-01 07:53:00	113
2024-01-01 07:00:00	2024-01-01 10:42:00	222
2024-01-01 08:00:00	2024-01-01 09:21:00	81
2024-01-01 09:00:00	2024-01-01 12:44:00	224

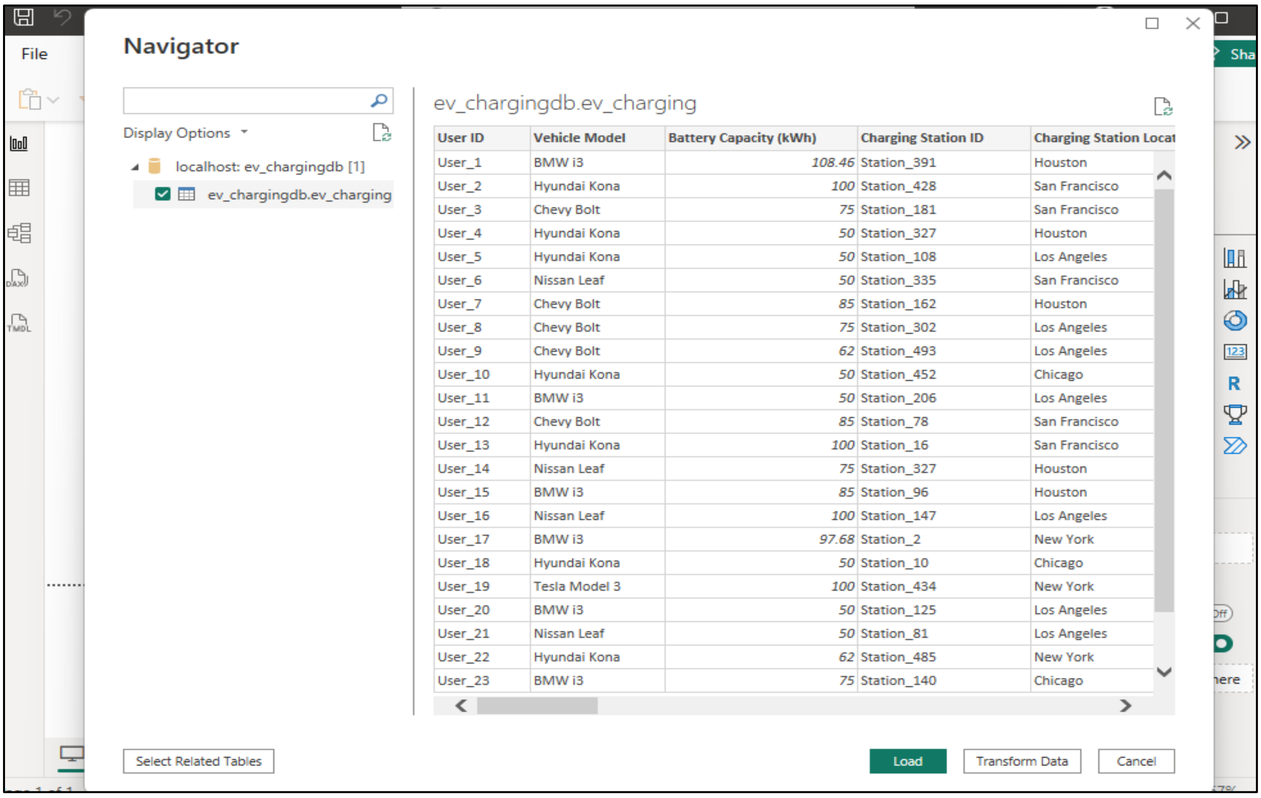
➢ Summary: After these steps the table contained typed datetime columns, a verified duration column, and numeric columns with invalid 0.0 values replaced by reasonable averages. The dataset was validated with SQL queries and prepared for import into Power BI for modelling and visualization.

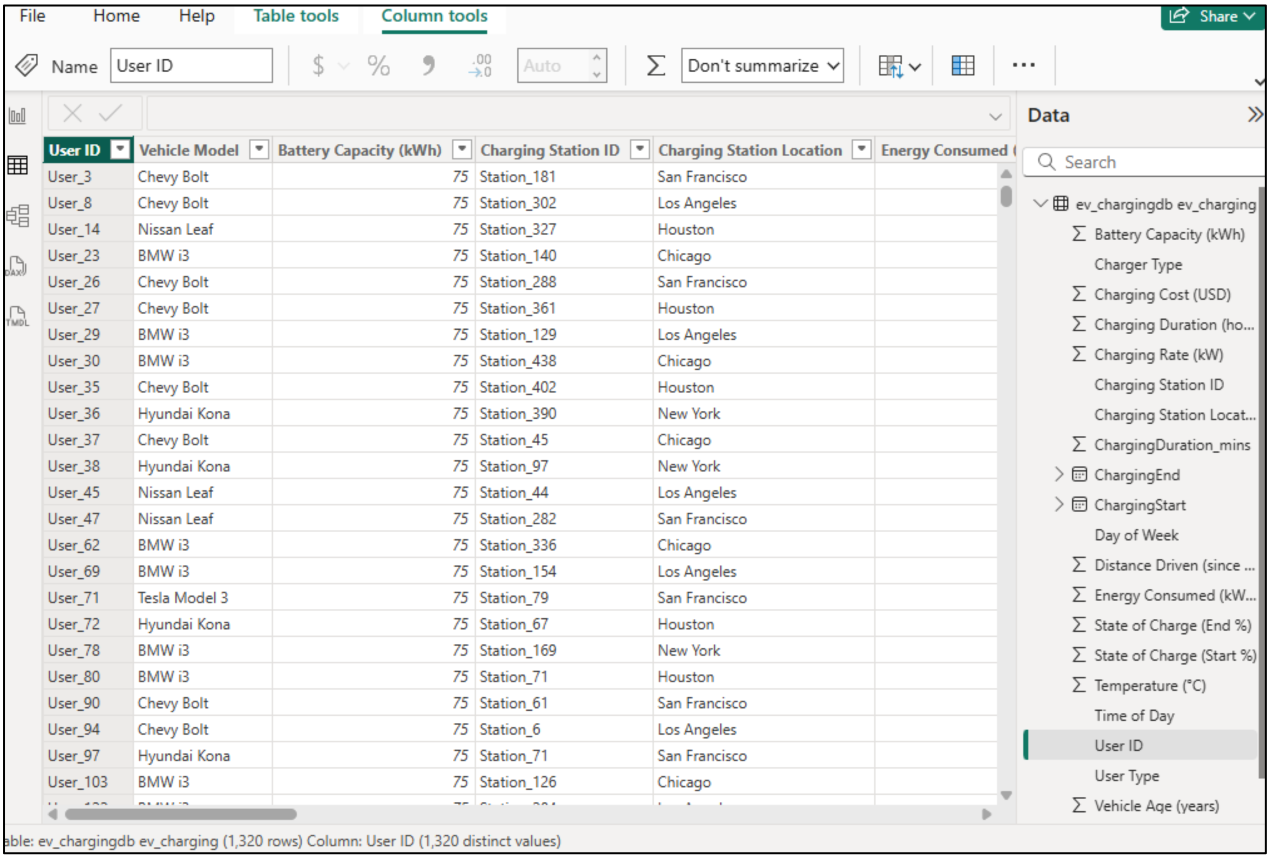
❖ **Power BI Data Modeling:**

➢ Database Connection- Connected Power BI Desktop to the MySQL Workbench database using the MySQL database connector. Selected the EV_Charging table which had been cleaned and validated in SQL.



➢ Data Import- Loaded the EV_Charging table into Power BI for further modeling. Verified that all important columns such as ChargingStart, ChargingEnd, ChargingDuration_mins, Energy Consumed and Charging Rate were imported correctly.





User ID	Vehicle Model	Battery Capacity (kWh)	Charging Station ID	Charging Station Location	Energy Consumed (
User_3	Chevy Bolt	75	Station_181	San Francisco	
User_8	Chevy Bolt	75	Station_302	Los Angeles	
User_14	Nissan Leaf	75	Station_327	Houston	
User_23	BMW i3	75	Station_140	Chicago	
User_26	Chevy Bolt	75	Station_288	San Francisco	
User_27	Chevy Bolt	75	Station_361	Houston	
User_29	BMW i3	75	Station_129	Los Angeles	
User_30	BMW i3	75	Station_438	Chicago	
User_35	Chevy Bolt	75	Station_402	Houston	
User_36	Hyundai Kona	75	Station_390	New York	
User_37	Chevy Bolt	75	Station_45	Chicago	
User_38	Hyundai Kona	75	Station_97	New York	
User_45	Nissan Leaf	75	Station_44	Los Angeles	
User_47	Nissan Leaf	75	Station_282	San Francisco	
User_62	BMW i3	75	Station_336	Chicago	
User_69	BMW i3	75	Station_154	Los Angeles	
User_71	Tesla Model 3	75	Station_79	San Francisco	
User_72	Hyundai Kona	75	Station_67	Houston	
User_78	BMW i3	75	Station_169	New York	
User_80	BMW i3	75	Station_71	Houston	
User_90	Chevy Bolt	75	Station_61	San Francisco	
User_94	Chevy Bolt	75	Station_6	Los Angeles	
User_97	Hyundai Kona	75	Station_71	San Francisco	
User_103	BMW i3	75	Station_126	Chicago	

- Calculated Columns in Power BI- Created additional calculated columns in DAX for analysis, which are as follows:

  - Hour of Day = FORMAT(ChargingStart, "HH")

```
Hour of Day = FORMAT ( 'ev_charging'[ChargingStart], "HH" )
```

  - Session Category = classified sessions into Short (30–59 mins), Medium (1–2 hours), Long (>2 hours)

```
Session Category = SWITCH( TRUE(),
    'ev_charging'[ChargingDuration_mins] >= 30 && 'ev_charging'[ChargingDuration_mins] < 60, "Short (30-59m)",
    'ev_charging'[ChargingDuration_mins] >= 60 && 'ev_charging'[ChargingDuration_mins] <= 120, "Medium (1-2h)",
    'ev_charging'[ChargingDuration_mins] > 120, "Long (>2h)")
```

<u>Report-1 page: Vehicle and User Analysis Report:</u>

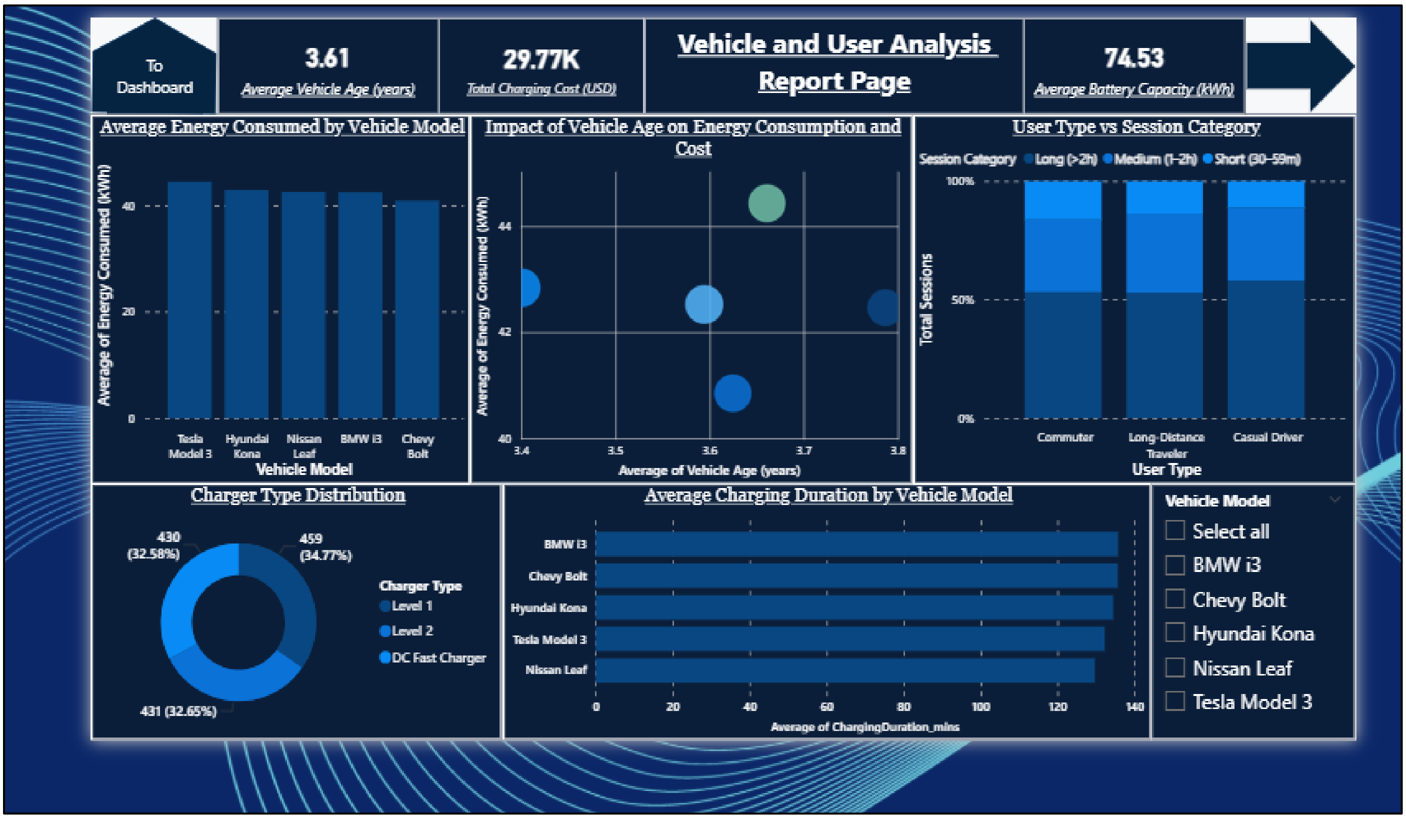
- This report provides a comprehensive analysis of charging patterns based on vehicle and user data, using interactive visualizations to deliver key insights.

- KPIs: The dashboard features key performance indicators (KPIs) at the top, including the Average Vehicle Age (years), Total Charging Cost (USD) and Average Battery Capacity(kWh). These metrics offer a quick, high-level overview of the dataset.
- Visualizations:

  - Bar chart: Displays the average energy consumed by each vehicle model.
  - Cluster column chart: Compares the average charging duration across different vehicle models.
  - Donut chart: Illustrates the distribution of charging costs by location, showing which areas have the highest total costs.

- Stacked column chart: Breaks down charging sessions by user type (Commuter, Long Distance Traveller, Casual Driver) and session length (Short, Medium, Long).
- Scatter plot: Analyze the relationship between vehicle age, energy consumption and charging cost.

➢ Slicers: Slicers for Vehicle Model, User Type and Charging Station Location provide flexibility, enabling users to drill down into the data and gain specific insights tailored to their needs.
➢ Navigation: The report includes a navigation pane with two buttons: one for the dashboard and another to navigate Report-2 page.



❖ **Report-2 page: Cost and Environmental Analysis Report:**

➢ This report page focuses on the financial and environmental aspects of EV charging, with a layout optimized for clarity and interactive exploration.
➢ KPIs: The dashboard displays three key metrics: Average State of Charge Gain (%), Average Temperature (°C) and Average Charging Cost per Session.
➢ Visualizations:
- Line Chart: Displays the monthly trend of charging sessions to track growth or decline in activity over time.
- Donut Chart: Illustrates the distribution of charging costs by location to show which areas have the highest total charging costs.
- Scatter Plot: Examines the effect of temperature on charging duration, plotting data to identify any correlation between environmental factors and charging time.
- Bar Chart: Highlights the top 10 charging stations by energy delivered to identify high-performing stations.
- Stacked Column Chart: Shows the distribution of charging sessions by day of the week and time of day, helping to identify peak usage patterns.

➢ Navigation: A navigation button is included to navigate to Report-1 page and Dashboard.
➢ Slicers: Slicers for Charger Type allowing users to dynamic selection and explore the data.

❖ KPIs of Dashboard:

1) Total Energy Consumed (kWh):

```
Session Category = SWITCH( TRUE(),
    'ev_charging'[ChargingDuration_mins] >= 30 && 'ev_charging'[ChargingDuration_mins] < 60, "Short (30-59m)",
    'ev_charging'[ChargingDuration_mins] >= 60 && 'ev_charging'[ChargingDuration_mins] <= 120, "Medium (1-2h)",
    'ev_charging'[ChargingDuration_mins] > 120, "Long (>2h)")
```

2) Total Charging Sessions:

```
Total Sessions = COUNTROWS ( 'ev_charging' )
```

3) Average Charging Rate (kW):

```
Average Charging Rate (kW) = AVERAGE ( 'ev_charging'[Charging Rate (kW)] )
```

4) Average Session Duration (mins):

```
Average Session Duration (minutes) = AVERAGE ( 'ev_charging'[ChargingDuration_mins] )
```

5) Utilization Percentage (%):

```
Utilization % = DIVIDE (
    COUNTROWS ( FILTER ( 'ev_charging', 'ev_charging'[Energy Consumed (kWh)] > 1 ) ),
    COUNTROWS ( 'ev_charging' ),
    0) * 100
```

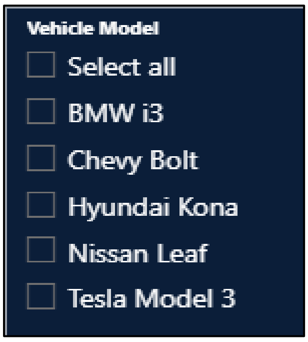
❖ Visualizations of Dashboard:

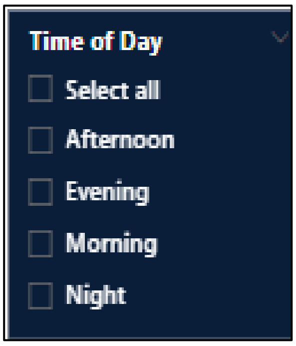
1) Line Chart – Charging Sessions Trend (Monthly): Shows how the number of charging sessions varied across January and February 2024. Helps identify growth or decline in session activity over time.
2) Donut Chart – Session Duration Distribution: Displays the proportion of short, medium, and long sessions. Useful for understanding which type of charging session dominates overall usage.
3) Bar Chart – Top 10 Charging Stations by Energy Delivered: Highlights the charging stations that delivered the most energy. Helps in identifying high-performing stations that contribute most to energy usage.
4) Scatter Plot – Distance vs Energy Consumed: Plots the relationship between distance driven and energy consumed. Shows a positive correlation indicating that higher distance usually requires more energy.
5) Stacked Column Chart – Charging Sessions by Day of Week and Time of Day: Breaks down the number of sessions across different days of the week and times of the day. Helps analyze peak usage patterns for planning and optimization.
6) Stacked Column Chart – Charging Sessions by Hour of Day and Day of Week: Displays the hourly distribution of sessions for each day of the week. Useful to identify peak charging hours and busiest days.

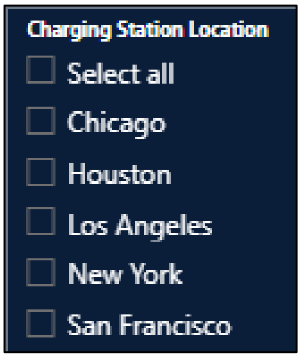
❖ Slicers of Dashboard:

➢ To make the dashboard interactive and user-friendly, the following slicers were added.

1) Vehicle Model: Enables filtering of charging sessions based on different vehicle models.
2) Time of Day: Segments charging sessions into Morning, Afternoon, Evening and Night. It helps analyze charging behaviour across different time periods.
3) Charging Station Location: Allows filtering of data by the location of the charging station. Useful for comparing charging patterns across different locations.







❖ Dashboard Design:

➢ The dashboard was designed to present information in a clear and interactive way. A consistent purple theme was applied, using darker shades for KPI cards and lighter tones for charts to improve readability.

1) Title and Subtitle:

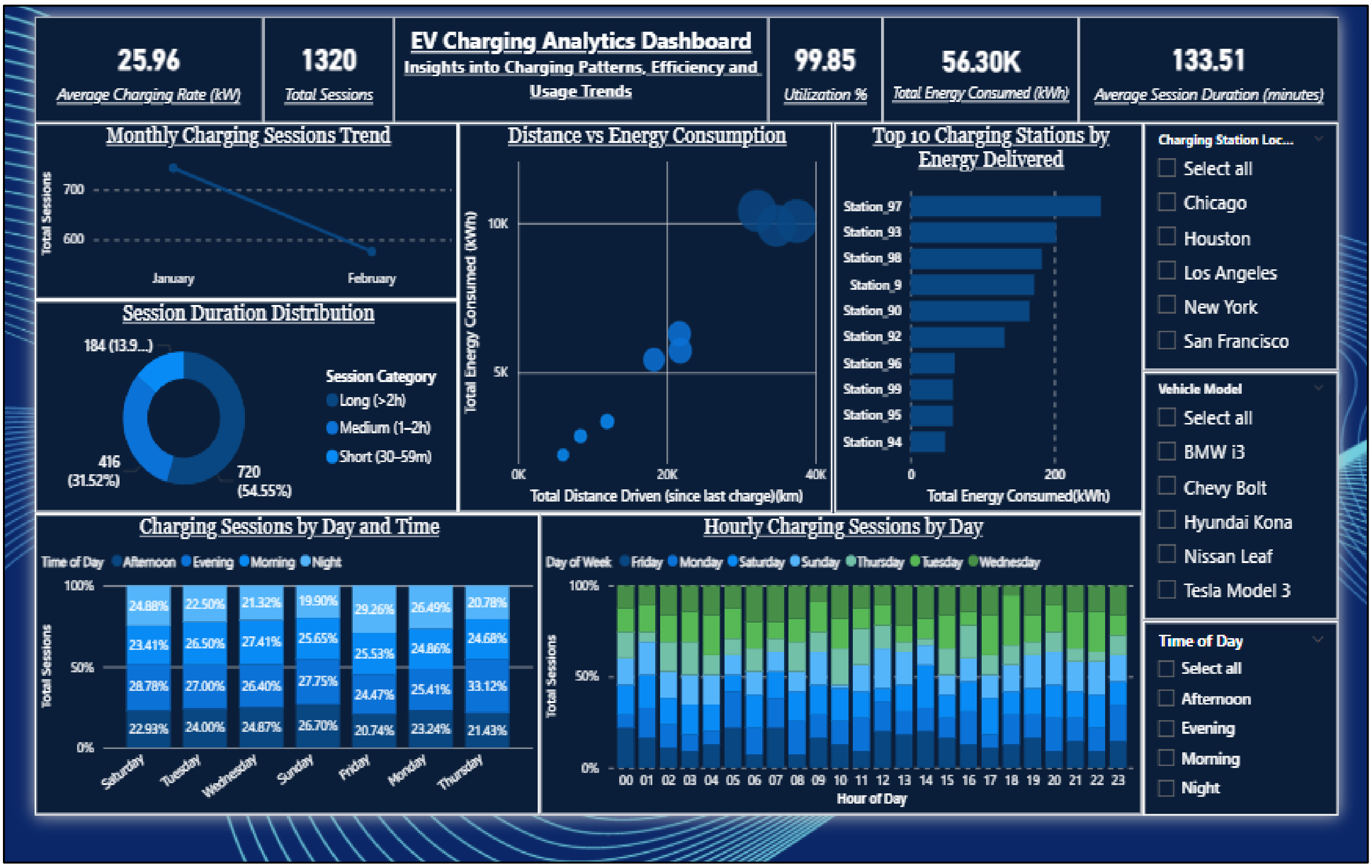
➢ The dashboard is titled "*Electric Vehicle Charging Pattern Analysis*" with the subtitle "*Insights into Charging Patterns, Efficiency and Usage Trends*".

2) Layout: The layout follows a structured arrangement:

- KPIs are placed at the top to provide quick access to key metrics.
- Charts are positioned in the center section to display trends, comparisons and usage patterns.
- Slicers are placed on the side panel to allow users to filter data dynamically by vehicle model, time of day and charging station location.

3) Dashboard Elements:

➢ In total, the dashboard contains 1 title, 5 KPIs, 6 charts and 3 slicers.



❖ **Key Insights:**

➢ The analysis of charging session data revealed several important patterns:

1) Peak usage was observed during evenings and weekends, indicating higher demand outside of working hours.
2) Medium-duration sessions (1–2 hours) made up the largest share of total charging activity, showing that most users prefer partial rather than full charges.
3) High-performing stations were identified, with the top 10 locations delivering the majority of energy consumed.
4) Distance and energy consumption showed a strong positive correlation, confirming that greater travel requires higher charging needs.
5) Utilization rate was nearly 100 percent, reflecting reliable data and consistent session activity.

❖ **Outcome:**

➢ The project successfully delivered an end-to-end data pipeline, beginning with raw CSV data and continuing through MySQL-based cleaning and transformation before visualizing results in Power BI. The outcome is a fully interactive dashboard that provides:

- Quick access to performance metrics through KPIs.
- Multiple visualizations for trends, comparisons and correlations.
- Flexible filters that allow users to explore data by vehicle model, time of day and station location.

➢ This outcome demonstrates how structured data engineering and business intelligence tools can generate actionable insights from real-world EV charging data.

❖ **Conclusion:**

- The Electric Vehicle Charging Pattern Analysis project demonstrates a complete workflow of data handling, transformation and visualization. Starting from CSV import into MySQL, the project applied SQL-based cleaning and added calculated fields before connecting the dataset to Power BI.
- The resulting dashboard provided clear insights into charging patterns, efficiency and infrastructure performance.
- This approach shows how electric vehicle charging data can be leveraged for planning and decision-making.
- The framework can be extended in the future to include larger datasets, real-time monitoring, predictive analytics and geospatial analysis for charging station placement.